

A 15-Qubit Recoherence Protocol for Benchmarking Mid-Circuit Measurements and Noise Models

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v49 was titled “A 15-Qubit Decoherence-Reversal Protocol for Benchmarking Mid-Circuit Measurements and Environmental Imprinting.” Changes are listed in the Revision Note.

Abstract Validating quantum processors at scale increasingly relies on approximate classical simulation, which adds truncation error to the comparison between ideal and observed behavior. We describe a 15-qubit dynamic-circuit benchmark that stays within exact state-vector simulation, so the ideal reference value is computed without approximation. The circuit scrambles a 6-qubit system, couples it to a 5-qubit engineered environment, reads out four weakly coupled probe qubits mid-circuit, and then applies the inverse of the scrambling and system–environment coupling to restore the system’s coherence. Exact summation over the 16 probe outcomes gives an ideal recovery fidelity $F_{\max} = 0.9889$ on the 11-qubit system–environment register; the shortfall from unity reflects information carried away by the probes, whose coupling is not inverted and which are discarded after readout. We demonstrate the pipeline with a single 400-shot run on the Quantinuum H2-1 emulator (`quantinuum.sim.h2-1e`) via Azure Quantum. A local surrogate built from three published H2-1 headline error rates predicted $F_{\exp} = 0.8353 \pm 0.0054$; the emulator returned a readout-mitigated $F_{\text{rec}} = 0.7836 \pm 0.0212$, 2.36 nominal standard errors lower. Because the surrogate’s quoted uncertainty reflects sampling error only, we interpret the shortfall as evidence of surrogate-model mismatch—expected, given the emulator’s richer noise model—rather than as a statistically calibrated anomaly. No physical-hardware data are reported.

1 Introduction

Large-scale demonstrations on quantum processors are validated against classical computation: cross-entropy benchmarking compares sampled bitstrings with simulated output distributions [1], and utility-scale experiments are compared with tensor-network simulations whose finite bond dimension introduces truncation error [2]. Dedicated benchmarks for mid-circuit measurements (MCMs) also exist, including randomized-benchmarking suites designed specifically for MCMs [3] and scalable mirror-circuit methods that, like the protocol here, use circuit inversion [4].

This note describes a narrower tool: a single fixed echo circuit with one mid-circuit measurement layer, small enough (15 qubits, 2^{15} amplitudes) that every measurement branch can be simulated exactly. Its purpose is to let three quantities be compared on the same circuit: an exact ideal value $F_{\max}$, a surrogate-noise expectation $F_{\exp}$, and an observed value F_{rec} . Restricting to an exactly simulable size removes approximation error from $F_{\max}$, so any gap between $F_{\exp}$ and F_{rec} reflects the noise model or the execution stack, not the reference calculation. The protocol complements, and does not replace, randomized-benchmarking approaches to MCM characterization.

1.1 Background and motivation

The protocol is motivated by questions from decoherence theory. In the einselection and Quantum Darwinism programs, classical behavior emerges as information

about preferred pointer states proliferates into the environment [5, 6]; see also Ref. [7] for a discussion of the measurement problem. Signatures of Quantum Darwinism have been observed in photonic cluster states [8], quantum jumps have been caught and reversed mid-flight using continuous monitoring and feedback [9], programmable scrambling has been realized and verified experimentally [10], and many-body echo circuits have been used on processors to characterize decoherence [11]. The present protocol does *not* test Quantum Darwinism: it does not measure the redundancy of records across environment fragments. It asks a simpler operational question—how much of an engineered unitary evolution can be reversed after information about the system has leaked into probe qubits that are read out mid-circuit and not uncomputed—and uses the answer as a benchmark.

2 Theoretical Framework

2.1 Closed-system idealization

Consider a system S interacting with an engineered environment E , starting in $|\psi_0\rangle_S|0\rangle_E$. Unitary interaction produces an entangled state [7]

$$|\Psi(t_1)\rangle_{SE} = \sum_k \alpha_k |\phi_k\rangle_S \otimes |E_k\rangle_E, \quad (1)$$

where $|\phi_k\rangle_S$ are pointer states. Tracing over E gives

$$\rho_S = \text{Tr}_E[|\Psi(t_1)\rangle\langle\Psi(t_1)|] \approx \sum_k |\alpha_k|^2 |\phi_k\rangle\langle\phi_k|_S. \quad (2)$$

If the joint evolution U is unitary and nothing is measured, applying $U^\dagger$ restores the initial state:

$$U^\dagger|\Psi(t_1)\rangle_{SE} = |\psi_0\rangle_S \otimes |0\rangle_E. \quad (3)$$

Undoing the system–environment entanglement in this way restores the coherence that ρ_S lost in Eq. (2), a process often called recoherence. This identity holds only for closed evolution in which every interaction is reversed. The protocol departs from it by coupling the system to a probe register that is not subsequently uncomputed; the probes are measured mid-circuit and discarded.

2.2 Echo with a mid-circuit measurement

In the standard Loschmidt echo, imperfect return arises because the reverse evolution differs from the forward one, $L(t) = |\langle\psi_0|e^{iH't}e^{-iHt}|\psi_0\rangle|^2$ with $H' \neq H$ [12]. Here, imperfect return arises because part of the forward evolution—the coupling to the probes—is not reversed, and the probe register is discarded.

Let the forward operation be $V = U_P U_{SE} U_S$: scrambling U_S on S , system–environment coupling U_{SE} , and weak system–probe coupling U_P . The probes are then measured in the computational basis, with projectors Π_m ($m \in \{0,1\}^4$). The implemented reversal is $R = (U_{SE}U_S)^\dagger$, which acts on $S \cup E$ only (on the full register, $R \equiv R_{SE} \otimes I_P$); the probe couplings are not inverted. The initial state is $\rho_0 = |\Psi_0\rangle\langle\Psi_0|$ with $|\Psi_0\rangle = |0\rangle_S^{\otimes 6} \otimes |0\rangle_E^{\otimes 5} \otimes |0\rangle_P^{\otimes 4}$. The unconditional final state is

$$\rho_f = \sum_m R \Pi_m V \rho_0 V^\dagger \Pi_m R^\dagger, \quad (4)$$

and the recovery fidelity, with the probe register traced out, is

$$\begin{aligned} F_{\text{rec}} &= \langle 0^{\otimes 11} | \text{Tr}_P(\rho_f) | 0^{\otimes 11} \rangle \\ &= \sum_m p_m |\langle 0^{\otimes 11}, m | R |\psi_m\rangle|^2, \end{aligned} \quad (5)$$

where $p_m = \|\Pi_m V |\Psi_0\rangle\|^2$ and $|\psi_m\rangle = \Pi_m V |\Psi_0\rangle / \sqrt{p_m}$. Evaluated for noiseless operations, Eq. (5) gives the ideal ceiling F_{max} .

Because R does not act on P and nothing is conditioned on the outcome m , the ideal reduced state is the same whether or not the probes are measured: with $\sigma = V \rho_0 V^\dagger$, $\text{Tr}_P(\sum_m \Pi_m \sigma \Pi_m) = \text{Tr}_P(\sigma)$. The ideal shortfall $1 - F_{\text{max}}$ is therefore set by the unreversed probe coupling and the loss of the information it transfers to the discarded probes, not by the act of measurement itself. The measurement matters operationally: on an emulator or device, performing a mid-circuit measurement brings its own error mechanisms (readout error, measurement crosstalk, dephasing of unmeasured qubits during readout), which recovery below F_{max} can expose. (Equation (4) of v49, $|\langle\psi_0|U^\dagger U|\psi_0\rangle|^2$, is identically 1 and has been replaced by Eqs. (4)–(5).)

Table 1: Protocol phases.

Phase	Action
I. Calibration	Define surrogate noise profiles
II. Exact simulation	F_{max} by branch summation
III. Forward	Scrambling and couplings
IV. Tracking	Mid-circuit probe readout
V. Execution	Submit OpenQASM 2.0 to H2-1E
VI. Analysis	Mitigate, compare with F_{exp}

3 Protocol

Table 1 summarizes the protocol phases.

3.1 Circuit specification

Qubits q_0 – q_5 form the system S , q_6 – q_{10} the environment E , and q_{11} – q_{14} the probes P .

1. **Scrambling** ($d = 15$ layers): each layer applies $R_y(\theta_{\ell,i})$ to each system qubit, with angles drawn uniformly from $[0, 2\pi)$ (NumPy `default_rng`, seed 42), followed by a CNOT ring $s_0 \rightarrow s_1 \rightarrow \dots \rightarrow s_5 \rightarrow s_0$.
2. **System–environment coupling**: CNOT $s_i \rightarrow e_i$ for $i = 0, \dots, 4$.
3. **Probe coupling**: controlled- $R_y(\lambda)$, $\lambda = 0.15$, from each control to its probe: $(s_0, s_1) \rightarrow p_0$, $(s_2, s_3) \rightarrow p_1$, $(s_4, s_5) \rightarrow p_2$, $(s_0, s_5) \rightarrow p_3$. The map was chosen so that every system qubit is coupled to at least one probe; it was not optimized.
4. **Mid-circuit measurement** of P .
5. **Reversal**: the S – E CNOTs in reverse order, then each scrambling layer inverted in reverse order. The probe couplings are not reversed.
6. **Final measurement** of S and E .

3.2 Exact baseline

F_{max} is computed by exact branch summation, Eq. (5), over all 16 probe outcomes on the full 2^{15} -dimensional state vector, giving $F_{\text{max}} = 0.9889$. In the ideal circuit, the probability that at least one probe registers a 1 is 0.031, and 2.1% of ideal recoveries occur with at least one probe click. We reproduced F_{max} with an independent implementation that does not use Qiskit.

3.3 Surrogate noise model

The circuit was transpiled for the H2-1E backend (Qiskit, optimization level 3) and simulated with Qiskit Aer using depolarizing error e_1 on single-qubit gates named **rx**, **ry**, **rz**, **sx**, **x**; depolarizing error e_2 on two-qubit gates named **cx**, **cry**, **cz**, **rzz**; and symmetric readout error e_{SPAM} on the target qubits. The “Typical” profile, $e_1 = 3 \times 10^{-5}$, $e_2 = 1 \times 10^{-3}$, $e_{\text{SPAM}} = 1 \times 10^{-3}$, uses published H2 headline specifications [13]. The “Pessimistic” profile, $e_1 = 2 \times 10^{-4}$, $e_2 = 2 \times 10^{-3}$, $e_{\text{SPAM}} = 5 \times 10^{-3}$, is a conservative choice made for this study, not a published bound. Each scenario uses 5,000 shots; quoted standard errors come from 200 bootstrap resamples and reflect sampling only, not uncertainty in the noise model itself.

Table 2: Sensitivity sweep (5,000 shots per scenario; standard errors from 200 bootstrap resamples, sampling only).

Scenario	d	λ	$(e_1, e_2, e_{\text{SPAM}})$	$F_{\text{clean}} \pm \text{SE}$	$F_{\text{exp}} \pm \text{SE}$	δ
H2 Typical (production circuit)	15	0.15	$(3 \times 10^{-5}, 10^{-3}, 10^{-3})$	0.9884 ± 0.0014	0.8353 ± 0.0054	0.1531
H2 Pessimistic (production circuit)	15	0.15	$(2 \times 10^{-4}, 2 \times 10^{-3}, 5 \times 10^{-3})$	0.9880 ± 0.0013	0.6805 ± 0.0071	0.3075
Deep Scramble (different circuit)	20	0.15	$(3 \times 10^{-5}, 10^{-3}, 10^{-3})$	0.9902 ± 0.0013	0.7831 ± 0.0064	0.2071
Strong Probe Coupling (different circuit)	15	0.30	$(3 \times 10^{-5}, 10^{-3}, 10^{-3})$	0.9556 ± 0.0029	0.8058 ± 0.0055	0.1498

3.4 Readout mitigation

Readout mitigation applies the inverse of a symmetric single-qubit assignment matrix, with error e_{SPAM} , as a tensor product over the 11 target bits; probe bits are ignored, implementing the partial trace in Eq. (5). Standard errors on mitigated values come from bootstrap resampling of the shot counts (400 resamples), with mitigation recomputed for each resample. The assignment matrix is treated as exact. The H2-1E documentation specifies asymmetric measurement errors (0.0006 following outcome 0, 0.00139 following outcome 1) [14], so the symmetric model is an approximation; at these error rates the mitigation correction is about 1%.

3.5 Execution pathway

In our tests, the standard Azure QIR submission path did not return correct per-shot results for this multi-shot circuit with mid-circuit measurements. We therefore compiled the circuit locally to OpenQASM 2.0 (`qasm2.dumps()`) and submitted it through the `Job.from_input_data()` API with Honeywell format tags. The classical registers were flattened into a single 15-bit register so that each shot returns one concatenated bitstring.

4 Results

4.1 Surrogate expectation

Table 2 lists the sensitivity sweep. F_{clean} is the noiseless shot-sampled estimate of F_{max} for each circuit; F_{exp} includes surrogate noise; $\delta = F_{\text{clean}} - F_{\text{exp}}$. Only the first two rows describe the production circuit. The Deep Scramble ($d = 20$) and Strong Probe Coupling ($\lambda = 0.30$; labeled “Strong QND Kick” in the v48 code) rows are different circuits that show sensitivity to depth and probe coupling; they are not alternative predictions for the production run.

4.2 Emulator run

A 400-shot job was run on `quantinuum.sim.h2-1e` (Job ID `cc0d204a-2346-11f1-8a26-0242ac1c000c`, submitted 19 March 2026 UTC). Readout mitigation of the target-register counts gives $F_{\text{rec}} = 0.7836 \pm 0.0212$. The archived output log lists only the ten most frequent bitstrings; among them, 304 shots had all 15 qubits reading 0. Under the mitigation model of Sec. 3.4, the reported F_{rec} corresponds to 310 shots (77.5%) in which the 11-qubit target register returned $|0\rangle^{\otimes 11}$; 309 or 311 such shots would give 0.7811 or 0.7861. The implied 6 recoveries with at least one probe click (1.9%) are close

to the 2.1% expected for the ideal circuit (Sec. 3.2); with six events this is descriptive only. (v49 reported a raw frequency of 76.0%; that figure is 304/400, the all-15-qubit ground-state frequency, not the target-register quantity defined in Eq. (5). F_{rec} is unaffected.)

4.3 Comparison with the surrogate

The shortfall relative to the Typical surrogate is $F_{\text{exp}} - F_{\text{rec}} = 0.0517$. With combined standard error $\sqrt{0.0054^2 + 0.0212^2} = 0.0219$, this is $z_{\text{nom}} = 2.36$ (one-sided $p \approx 0.009$). This statistic is *conditional on the Typical surrogate being the correct predictive model*: it includes shot and Monte Carlo sampling error but no uncertainty in the noise model, the readout calibration, or run-to-run variation. The result exceeds a nominal 2σ criterion but not a 3σ criterion, and the 2σ threshold was not fixed in advance. The observed value also lies between the Typical (0.8353) and Pessimistic (0.6805) predictions for the same circuit.

The v48 analysis code labels this outcome “Anomalous Decoherence.” We now describe it as a *nominal surrogate-model mismatch*: the observed recovery is lower than a three-parameter surrogate predicts, by more than that surrogate’s sampling error alone can explain.

5 Discussion

The mismatch is expected. The H2-1E emulator implements a detailed noise model that includes asymmetric measurement errors, measurement and initialization crosstalk, initialization faults, spontaneous emission, linear and quadratic dephasing with coherent and incoherent components, and angle-dependent two-qubit error scaling [14]. The local surrogate contains none of these beyond headline depolarizing and readout rates, so a lower emulator recovery is unsurprising. What the run demonstrates, on a single fixed circuit and 400 shots, is a measurable discrepancy between the headline-specification surrogate and the full-stack emulator.

Version 49 attributed the gap partly to SWAP gates inserted during routing. That explanation is withdrawn. System Model H2 is a quantum charge-coupled device (QCCD) trapped-ion processor that physically transports ions between gate zones [15], so logical SWAP insertion is not the expected mechanism. Transport- and memory-time-dependent dephasing may contribute, but this experiment cannot separate them from the other mechanisms listed above.

The protocol reverses the scrambling and system–

environment coupling but not the probe coupling. The ideal ceiling $F_{\max} = 0.9889$ therefore reflects information carried away by the discarded probes ($1 - F_{\max} \approx 0.011$). Absent execution-path discrepancies, which the noiseless-emulator control in Sec. 6 would rule out, recovery below $F_{\max}$ is attributable to noise introduced by the emulator and execution stack, whether modeled or not.

6 Limitations and Next Steps

- **Single run.** One 400-shot emulator job; no repeated runs, so run-to-run variation is unmeasured. Only the ten most frequent bitstrings were archived, so the target-register count is inferred from F_{rec} rather than read from the full shot record.
- **No hardware data.** All results come from an emulator. No claim is made about physical decoherence on H2-1.
- **No noiseless-emulator control.** Running the same compiled circuit on H2-1E with its error model disabled (`error-model: False` [16]) would verify serialization, register handling, and mid-circuit measurement semantics end to end. This control was not performed.
- **Uncalibrated decision rule.** The 2σ threshold was not preregistered, and the comparison omits surrogate-model and calibration uncertainty. A predictive variance including these terms is needed before any classification rule can be interpreted as a false-positive rate.
- **Readout model.** Mitigation assumes symmetric, uncorrelated readout errors with an exactly known assignment matrix.
- **Single circuit instance.** One fixed random seed; no averaging over circuit instances.

A stronger version of this study would add the noiseless-emulator control, repeat the noisy run, propagate surrogate-parameter uncertainty, and then test the same circuit on hardware with a decision rule fixed in advance.

Data and Code Availability

Code (local simulation and Azure submission scripts, v48), results log, and manuscript: <https://github.com/dkahng/15-qubit-mcm-protocol> (MIT License). Archived record: Zenodo, DOI 10.5281/zenodo.19105460. Also listed on OSF (<https://osf.io/9dnm8>) and ResearchGate.

Revision Note (v50)

- Retitled; abstract and discussion rewritten to describe the result as surrogate-model mismatch on an emulator, not anomalous decoherence.
- Eq. (4) of v49, which is identically 1, replaced by a channel formulation including the mid-circuit measurement (Eqs. 4–5).
- Raw ground-state frequency corrected from 76.0% (304/400, all 15 qubits) to 77.5% (310/400, target register, inferred from the reported F_{rec}).
- Statistical comparison restated as a conditional nominal

z -score; limitations of the 2σ rule stated.

- The ideal shortfall $1 - F_{\max}$ is attributed to the unreversed probe coupling and discarded probes rather than to measurement backaction; ρ_0 defined explicitly.
- SWAP-gate explanation withdrawn.
- Circuit, surrogate, and mitigation details added; Table 2 now lists d and λ and identifies which rows describe the production circuit.
- References: Blais et al. removed; Kim et al. and Blok et al. now cited only for claims they support; added Gorin et al., Govia et al., Proctor et al., Moses et al., and Quantinuum and Microsoft documentation.
- Data and code availability statement added.

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